Pre-Processing of Remotely Sensed Data

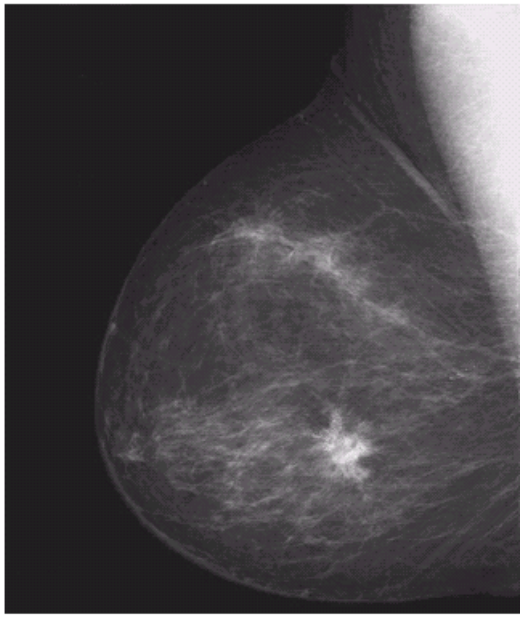
Point Processing -Image Negatives

- Negative of an image with gray levels in the range [0,L-1]is obtained by using the negative transformation
    - S = (L – 1) - r
  - values of pixels before Processing = r
  - values of pixels after processing = S
- Reversing the intensity levels of an image in this manner produces the equivalent of a photographic negative
- This type of processing , Negative images are useful for enhancing white or grey detail embedded in dark regions of an image

Point Processing: Negative Images

– Note how much clearer the tissue are in the negative image

Original Image





Negative Image

Brightness Inversion



Inverted shades helps certain features stick out more that were hard to see before

Histogram Matching



Histogram Matching

- Modifies the histogram of one image so that it resembles the histogram a second image
- Histogram matching is useful for matching data of the same or adjacent scenes that were acquired on separate days, or are slightly different sun angle or atmospheric effects
- Improves display
- Mosaicking (images of different time, illumination condition, contrasts)
- Histogram matching is performed band-to-band. For example, band 2 of reference image is matched to band 2 of the input image
- Number of bands of both images should be equal

CONTRAST ENHANCEMENT

- Spreads the brightness values over a broader range, allowing the eye to see detail formerly concealed in the extremely dark or bright tones
- Detecting and emphasising detail in an image